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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

BT2203-1 - FOOD BIOTECHNOLOGY
(For BT)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Fivefull** questions choosing **Two full** question from each **Unit – I and II**, **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. State the nutritional aspects of food.
A) Only water content
B) Only carbohydrates
C) Energy, macronutrients, micronutrients
D) Only minerals
2. Choose a major role of microorganisms in food.
A) Freezing
B) Packaging
C) Color addition
D) Food spoilage and fermentation
3. When did microbes first become recognized in food?
A) After GM food development
B) Only in 20th century
C) During refrigeration era
D) Ancient history of food spoilage
4. Which type of microorganism is used in alcoholic beverage production?
A) Bacteria
B) Yeasts
C) Molds
D) Algae
5. Identify a bacterial foodborne disease.
A) Measles
B) Influenza
C) Salmonellosis
D) HIV
6. How does high temperature preserve food?
A) Destroys microorganisms
B) Freezes food
C) Ferments food
D) Enhances bacterial growth
7. What is the effect of freeze drying on microorganisms?
A) Preserves food by reducing water activity
B) Kills all microbes
C) Increases microbial growth
D) Adds nutrients
8. Which natural preservative inhibits microbial growth in food?
A) Salt
B) Sugar
C) Water
D) Alcohol-free vinegar
9. Why are lipases used in food processing?
A) Hydrolyze proteins only
B) Hydrolyze fats and improve texture
C) Reduce color
D) Enhance flavor only
10. Which functional food is rich in health-promoting compounds?
A) Cake
B) White rice
C) Bread
D) Golden rice
11. How do enzymes improve food manufacturing?
A) Only packaging
B) Only color enhancement
C) Reduce spoilage, enhance flavor, and improve shelf life
D) Only heating

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12. Identify an example of prebiotics.

- A) Corn starch
B) Lactobacillus
C) Yeast
D) Inulin

13. Which technique uses most probable number for microbial analysis?

- A) Direct plate count
B) Statistical estimation of bacteria in liquid samples
C) ELISA
D) DNA sequencing

14. What is a rapid immunological method for microbial detection?

- A) Standard plate count
B) Fluorescent antibody technique
C) Direct microscopic count
D) Drying

15. Which factor primarily accelerates food spoilage?

- A) Low humidity
B) Refrigeration
C) Vacuum packaging
D) High moisture content

16. Why is electrical method used in microbial detection?

- A) To detect microbial activity rapidly
B) To kill microbes
C) To preserve food
D) To measure flavor

17. What is monosodium glutamate used for in food?

- A) Flavor enhancement
B) Sweetener
C) Preservative
D) Coloring agent

18. Single Cell Protein is best described as:

- A) Protein obtained from animals
B) Protein obtained from cereals
C) Protein obtained from meat only
D) Protein obtained from single-celled microorganisms

19. How is kimchi classified as a novel food?

- A) Alcoholic beverage
B) Dairy product
C) Fermented vegetable product
D) Processed cereal

20. Identify a dairy product that relies on lactic acid fermentation

- A) Cheese
B) Beer
C) Wine
D) Bread

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Compare traditional food processing methods with modern food biotechnology, highlighting the role of rDNA technology.	8	L*2	1	2
	b) Outline the symptoms, modes of transmission, and preventive measures of Norovirus and Rotavirus mediated food-borne infections.	8	L2	2	2
2.	a) Summarize the impact of microbial spoilage of fruits, vegetables, and meat products.	8	L2	1	1
	b) Discuss the major diversity of <i>Salmonella</i> with respect to species, and serotypes. Add a note on their pathogenicity, and preventive strategies.	8	L1	2	2
3.	a) List the advantages and limitations of natural food preservatives, such as salt, sugar, and vinegar.	8	L1	2	1
	b) Summarize the role and importance of microorganisms in food, providing examples of their applications in food production and processing.	8	L2	1	1
Unit – II					
4.	a) Illustrate the applications of hydrolases in food manufacture.	8	L1	3	3
	b) List the parameters of food that influence microbial growth.	8	L2	4	2,3

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| 5. | a) | Summarize the benefits of probiotics and prebiotics in food. | 8 | L1 | 3 | 2 |
| | b) | Demonstrate the dye-reduction test to assess microbial load in food.
Add a note on the methodological approach. | 8 | L2 | 4 | 3 |
| 6. | a) | Interpret HACCP as a tool for ensuring microbial food safety. | 8 | L2 | 4 | 3 |
| | b) | Outline the regulatory aspects of genetically modified foods. | 8 | L2 | 3 | 3 |

Unit – III

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|----|----|--|---|----|---|---|
| 7. | a) | Summarize the significance of single-cell proteins (SCP) like Spirulina in terms of nutrition, health, and sustainability. | 8 | L2 | 5 | 1 |
| | b) | How is lactic acid produced by microorganisms, and what are its uses? | 8 | L2 | 5 | 1 |
| 8. | a) | Discuss the contribution of microorganisms to the production and quality of cereal-based foods like bread. | 8 | L2 | 5 | 1 |
| | b) | Define food additives and illustrate their classification with common examples. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
